

Platform Engineering 101: Building Internal Developer Platforms

Maarten

Red Hat

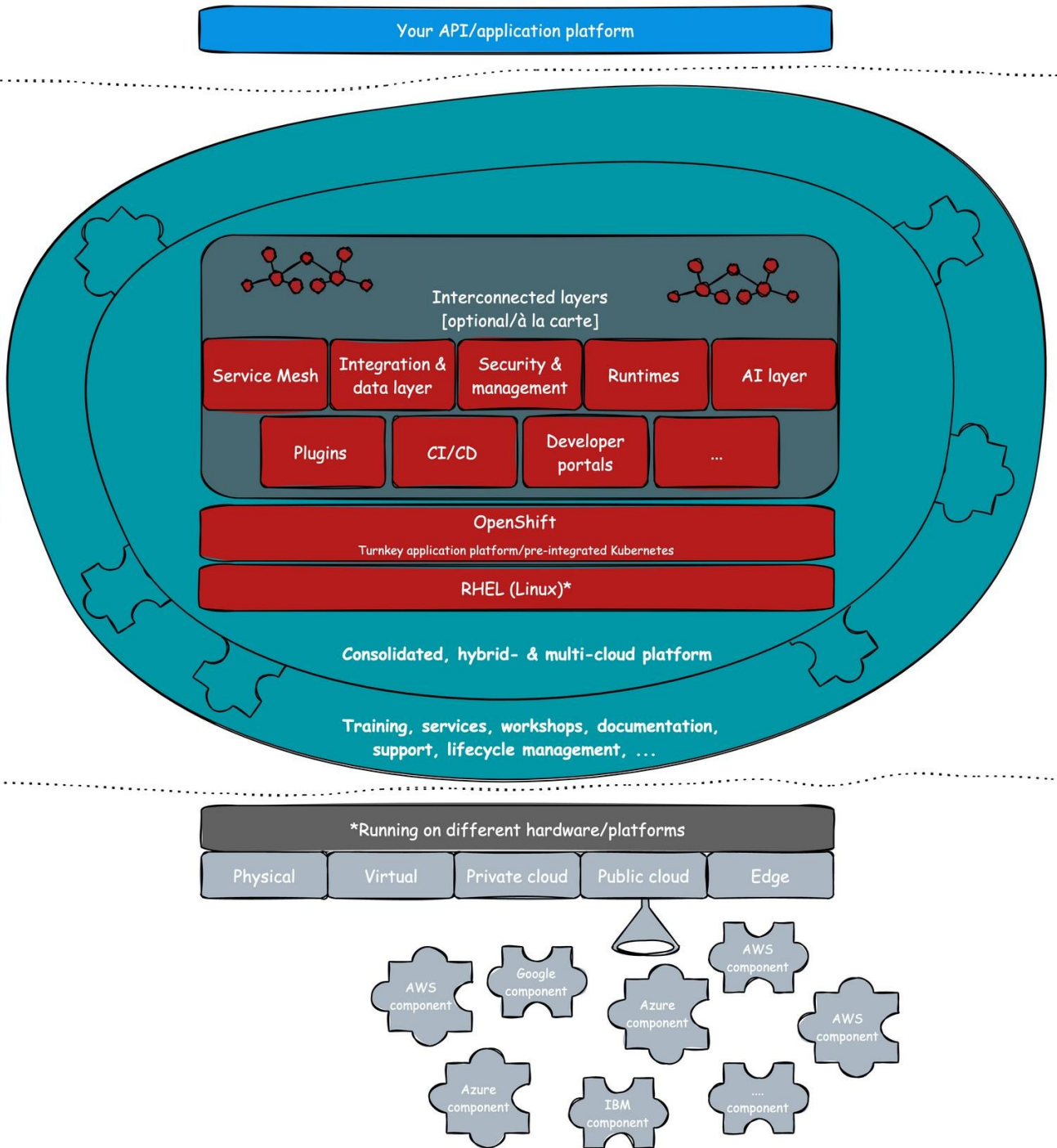
Maarten Vandepierre

maarten-vandepierre.be

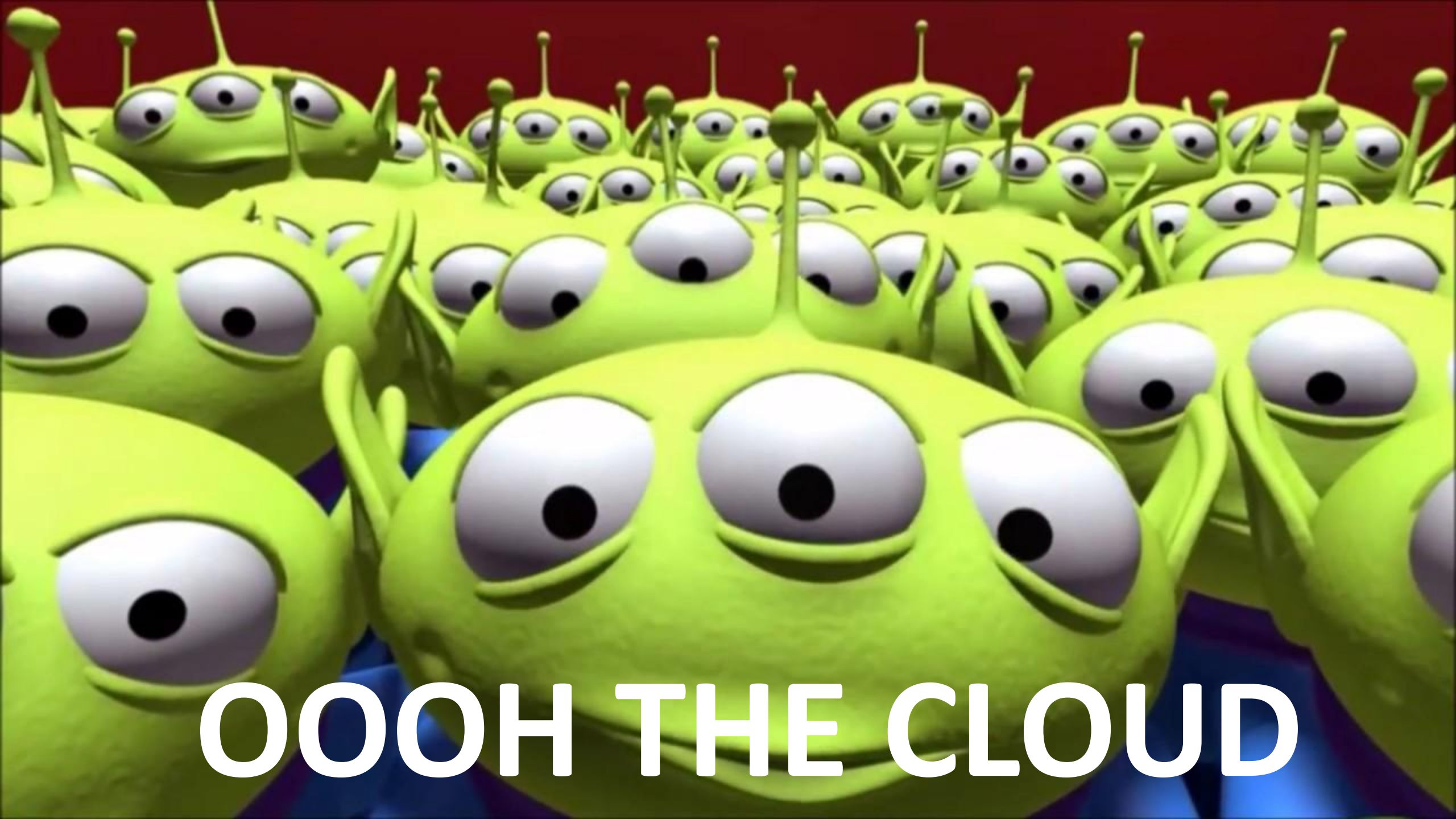
Maarten Vandepierre



Red Hat







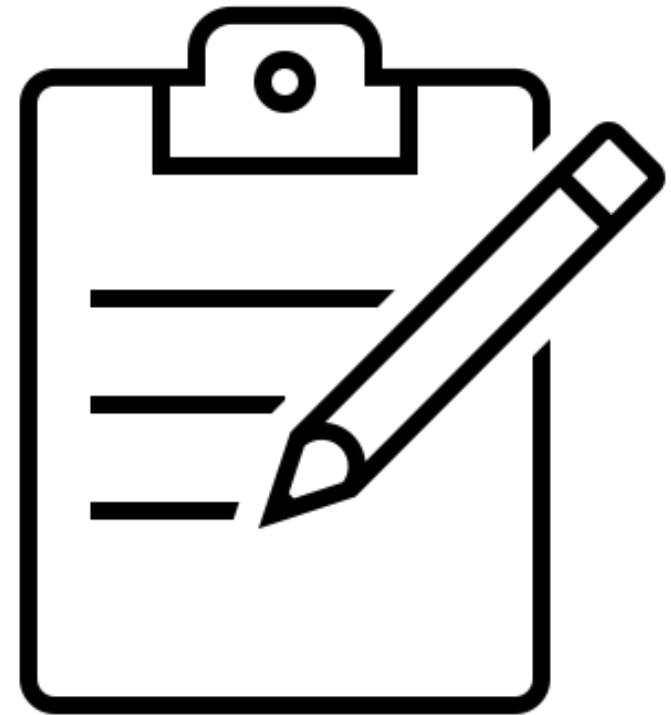
OOOH THE CLOUD



$$\begin{aligned} H_0 &= \mu_1 - \mu_2 - \mu_3 - \dots \\ H_1 &= \mu_1 \sum \langle i | \psi \rangle^2 \\ H_2 &= \mu_1 \sum \langle i | \psi \rangle^2 \\ &\quad + \mu_2 \sum \langle i | \psi \rangle^2 \\ H_3 &= \mu_1 \sum \sum \sum \langle i | \psi \rangle^2 \\ &\quad + \mu_2 \sum \sum \langle i | \psi \rangle^2 \\ &\quad + \mu_3 \sum \langle i | \psi \rangle^2 \\ G_7 &= P_7 LS \end{aligned}$$

Agenda

- How Did We Get Here?
- Developer Struggles
- Platform Engineering
- Internal Developer Platform (IDP)
- Internal Developer Portal
- Benefits for Developers
- Tools to Help
- **Demo (25-30 min!!)**
- Summary & Resources



What
Happened?!



Shift Left:
Increasing
cognitive
load



It's not just
app development
anymore

Dev

Ops

DevOps

FinOps

MLOps

GitOps

DataOps

DevSecOps

BizDevOps

BizDevSecOps

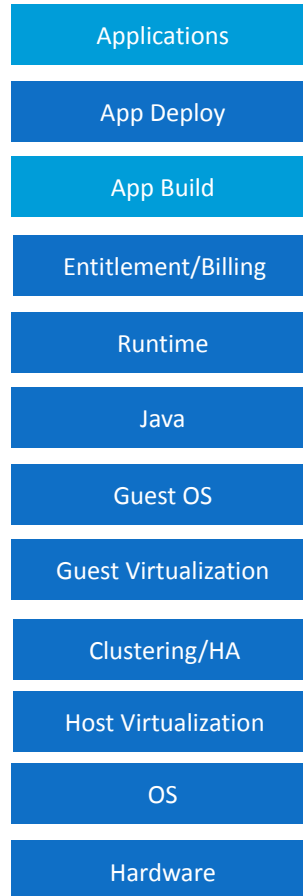
BizDevSecFinMLOps

Organizational Impact & Cognitive Load

Traditional App Delivery



Dev



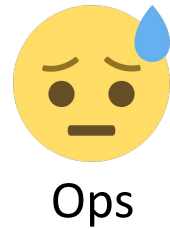
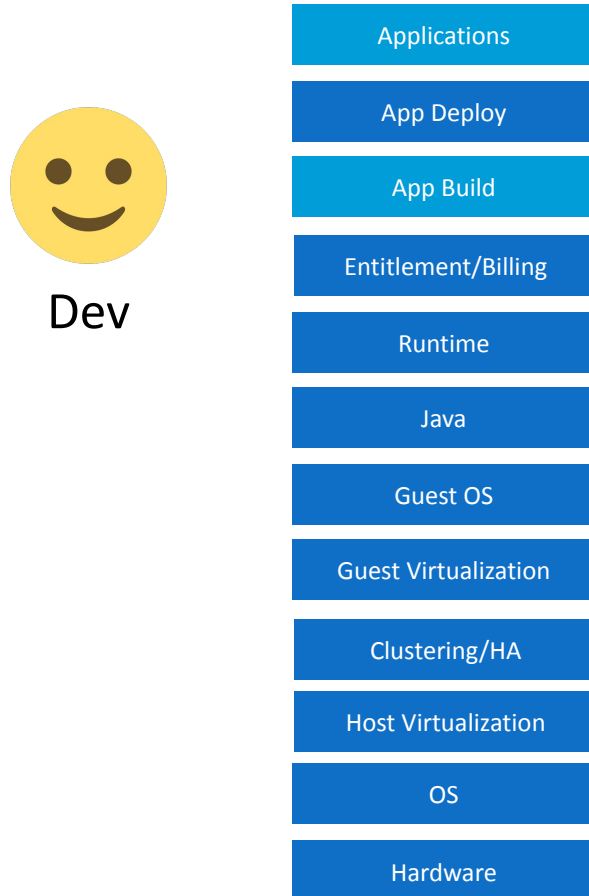
Ops

Key:



Organizational Impact & Cognitive Load

Traditional App Delivery

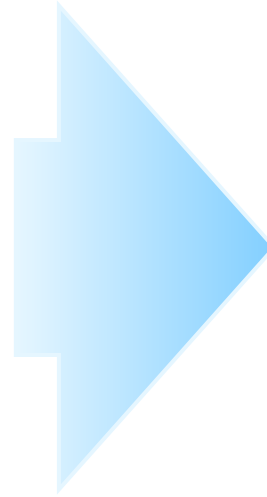


Ops

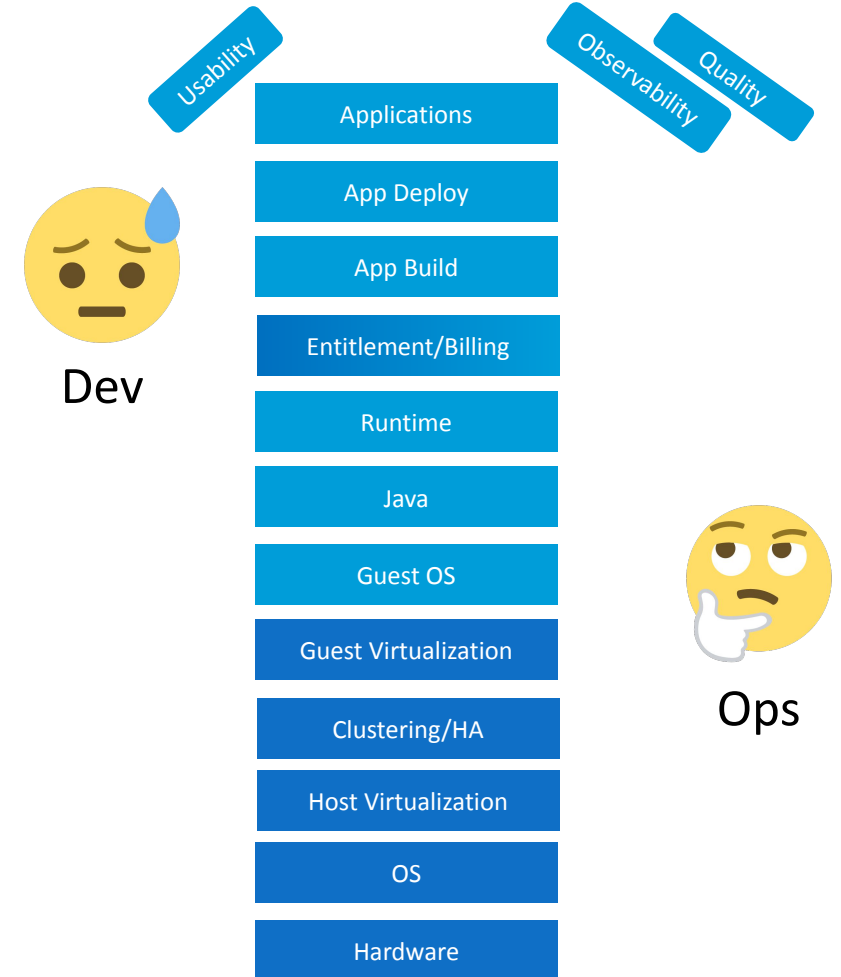
Key:

Dev Responsible

Ops Responsible



Modern App Delivery

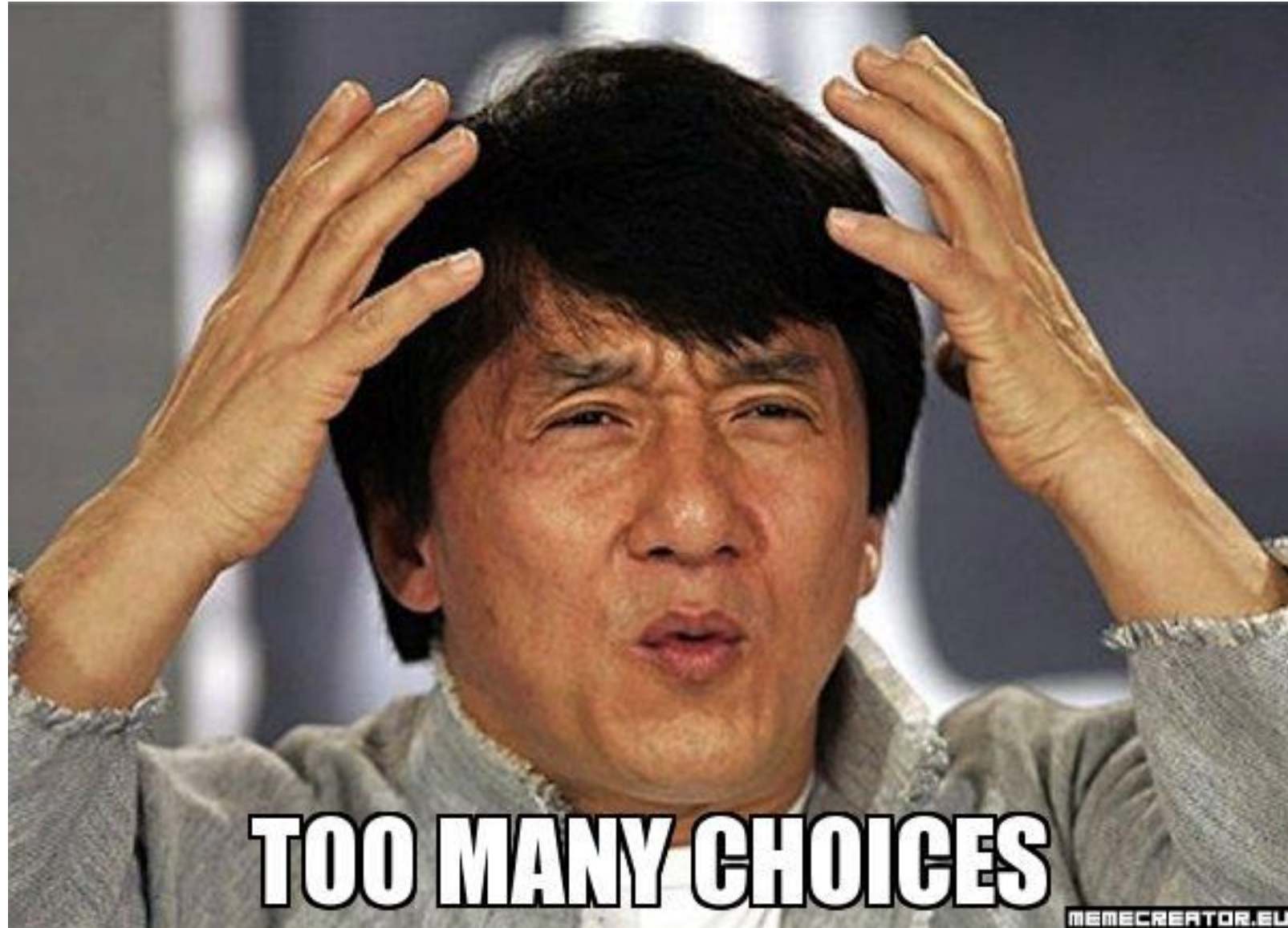


Dev



Ops

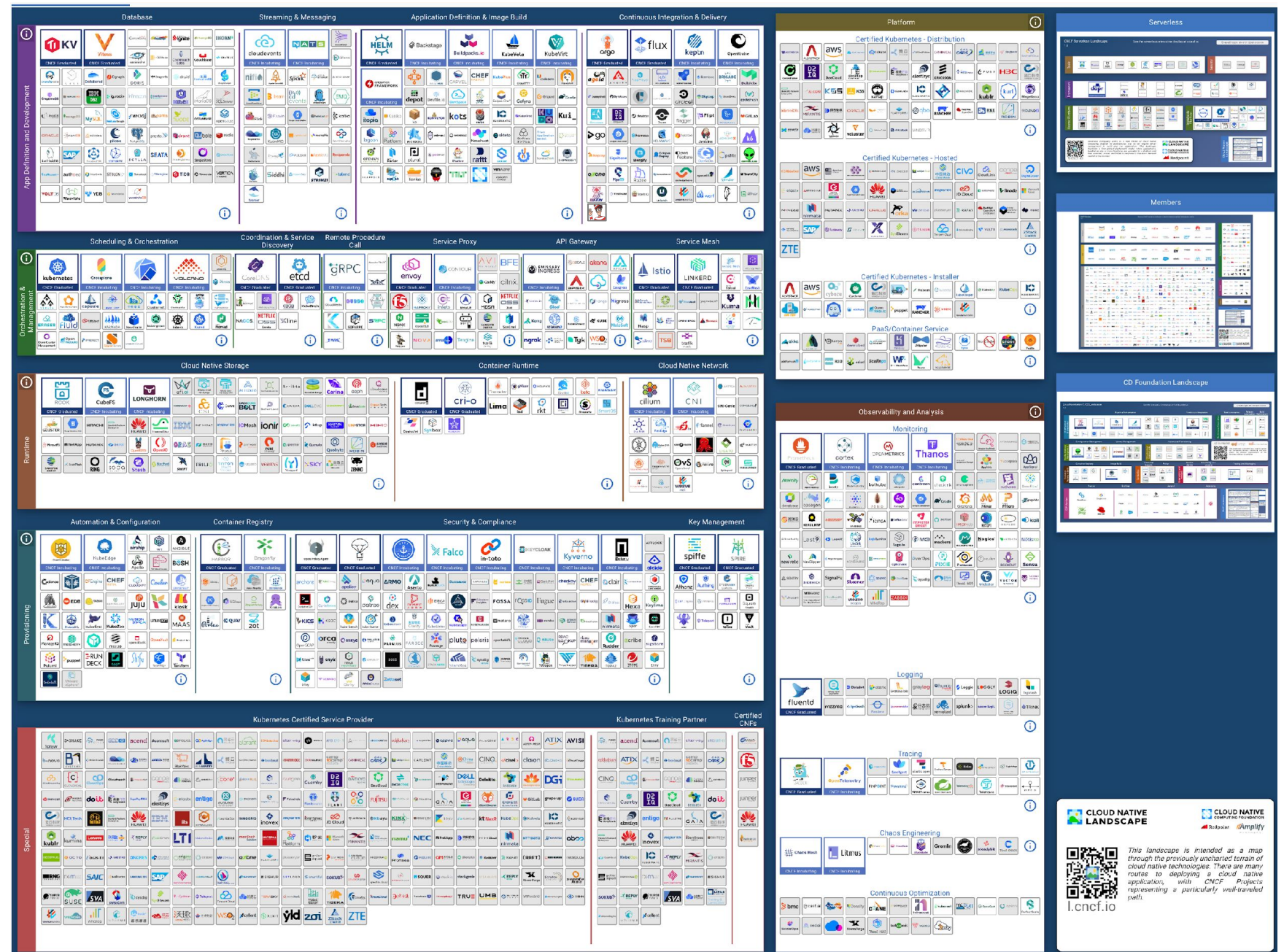
Tools:
Too much
choice



The Cloud-Native Grocery Store

Where's Waldo?

or Wally?

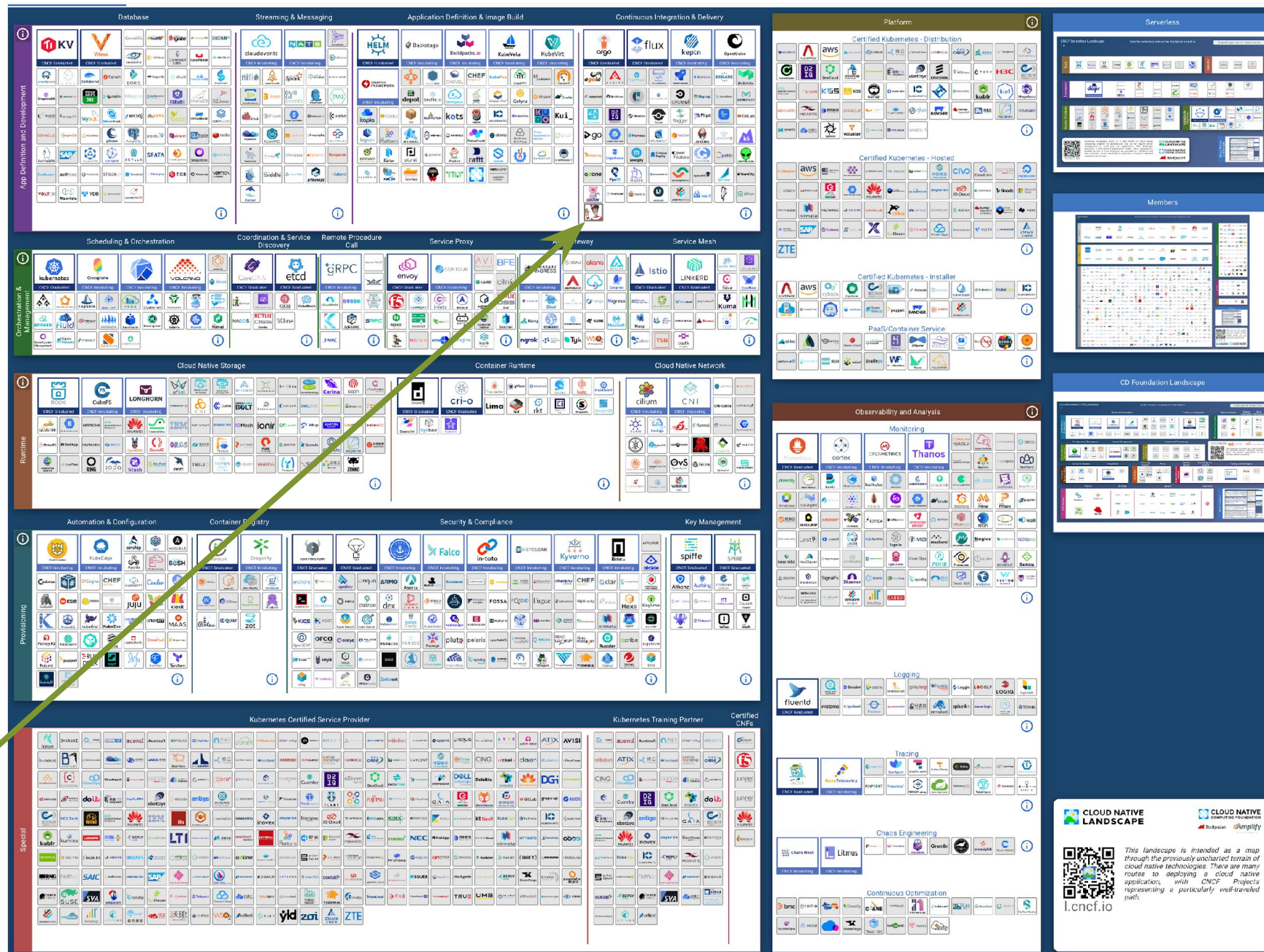


Source: <https://landscape.cncf.io/>

The Cloud-Native Grocery Store

Where's Waldo?

or Wally?



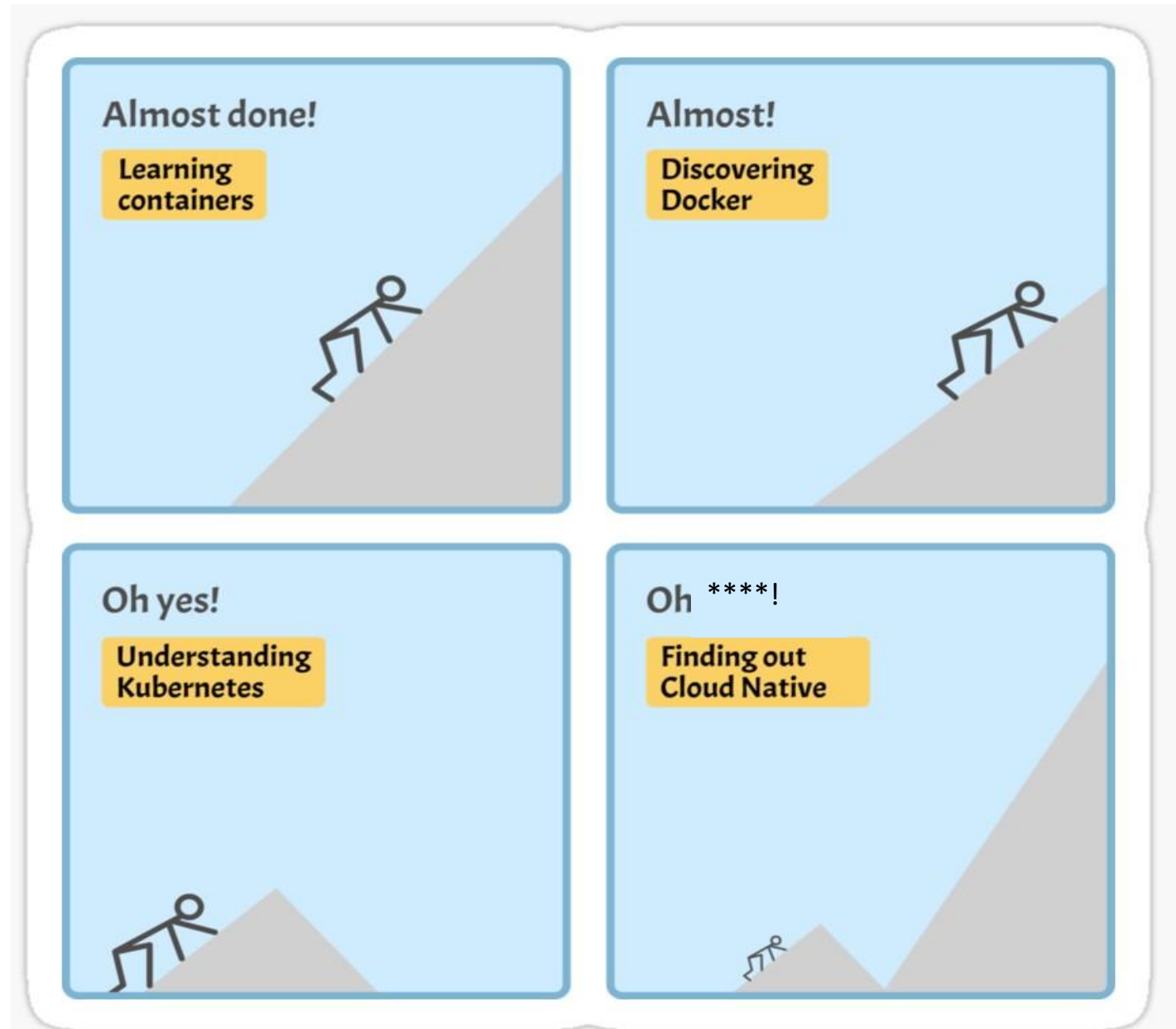
Source: <https://landscape.cncf.io/>

Developer Struggles



1. CI/CD Tooling

- Steep learning curve – takes time
- Too many options
- OSS standards – still too many!
- Observability



Source: https://www.reddit.com/r/ProgrammerHumor/comments/lgmj45/adopting_cloud_native/

2. Container Management

- Offers portability and flexibility
- Many decisions needed
 - Which tools to use?
 - How to configure environments?
 - How to build our apps?
- Challenging to remain consistent and stick to best practices
- Operators & Cloud Native Buildpacks can help



Source: <https://semaphoreci.com/blog/kubernetes-deployment>

3. Security & Regulatory Requirements

- Compliance & regulatory awareness
- SBOM –what's in your application?
- Vulnerability awareness
 - Source code
 - Images
 - External libraries
- Trusted software supply chain
- Compromised credentials?
- Authentication & Authorization



Source: <https://www.asylas.com/asylas/security-memes-our-year-end-humor-break/>

Other Struggles

- Likely drowning in scripts and maintenance costs
- Can't reach degree of automation wanted
- Key-person dependency
- Creating a new environment can be loooooong
- Environments can get blocked testing



What developers are saying....

Top three causes of developer burnout:

1. Increasing workload/demand from other teams (39%)
2. Pressure of digital transformation (37%)
3. Requirement to learn new skills (35%)

76% of organizations say cognitive load is so high it causes angst and low productivity for developers.



Luca ✓ @luca_cloud · Aug 24, 2022

Devs... be real with me. Do y'all want to do Ops? 🧵

Yes

41.8%

No

20.3%

Fuck No

21.8%

I don't care 🙄

16.1%

330 votes · Final results

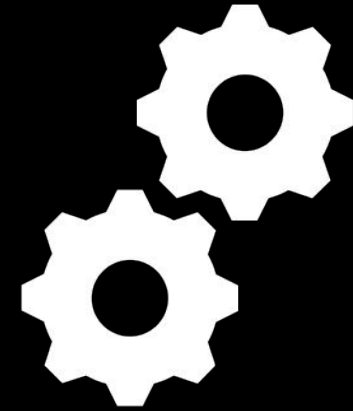
Replying to @luca_cloud

https://twitter.com/intent/tweet?ref_src=twsrc%5Etfw%7Ctwcamp%5Etweetembed%7Ctwterm%5E1562349679660122112%7Ctwgr%5Ea1005affb9784baa8ed6ee07c455995f8ad267df%7Ctwcon%5Es1_&ref_url=https%3A%2F%2Fblog.gitguardian.com%2Fplatform-engineering-and-security-a-very-short-introduction%2F&in_reply_to=1562349679660122112

But... How to overcome
these struggles?



Platform Engineering



What is Platform Engineering?

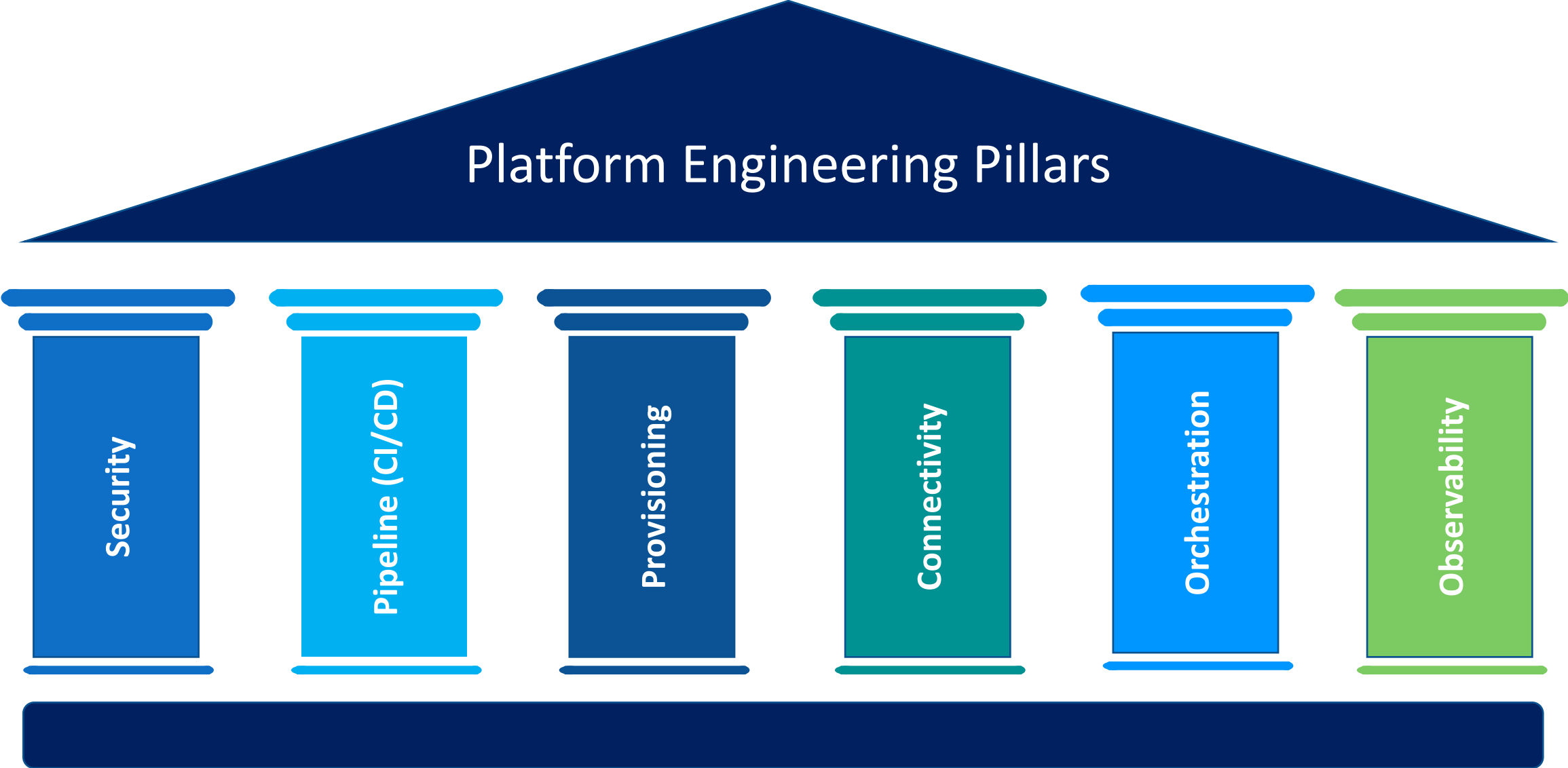
Platform engineering is the discipline of designing and building toolchains and **workflows** that enable **self-service capabilities** for software engineering organizations in the cloud-native era. Platform engineers provide an integrated product most often referred to as an “**Internal Developer Platform**” covering the operational necessities of the entire lifecycle of an application.

- platformengineering.org



Source: <https://medium.com/defense-unicorns/5-minute-devops-platform-engineering-doesnt-replace-devops-599f2d61b66>

Platform Engineering Pillars

The diagram illustrates the six pillars of Platform Engineering as a classical building. A dark blue triangular pediment at the top contains the title 'Platform Engineering Pillars'. Below it are six vertical pillars of varying heights and colors, each with a label. From left to right, the pillars are: Security (dark blue), Pipeline (CI/CD) (light blue), Provisioning (dark blue), Connectivity (teal), Orchestration (bright blue), and Observability (green). Each pillar has a decorative top and bottom flange. A solid dark blue horizontal base runs across the bottom of the pillars.

Security

Pipeline (CI/CD)

Provisioning

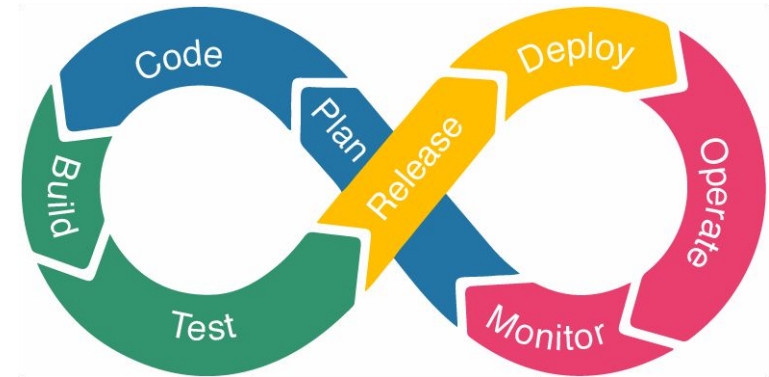
Connectivity

Orchestration

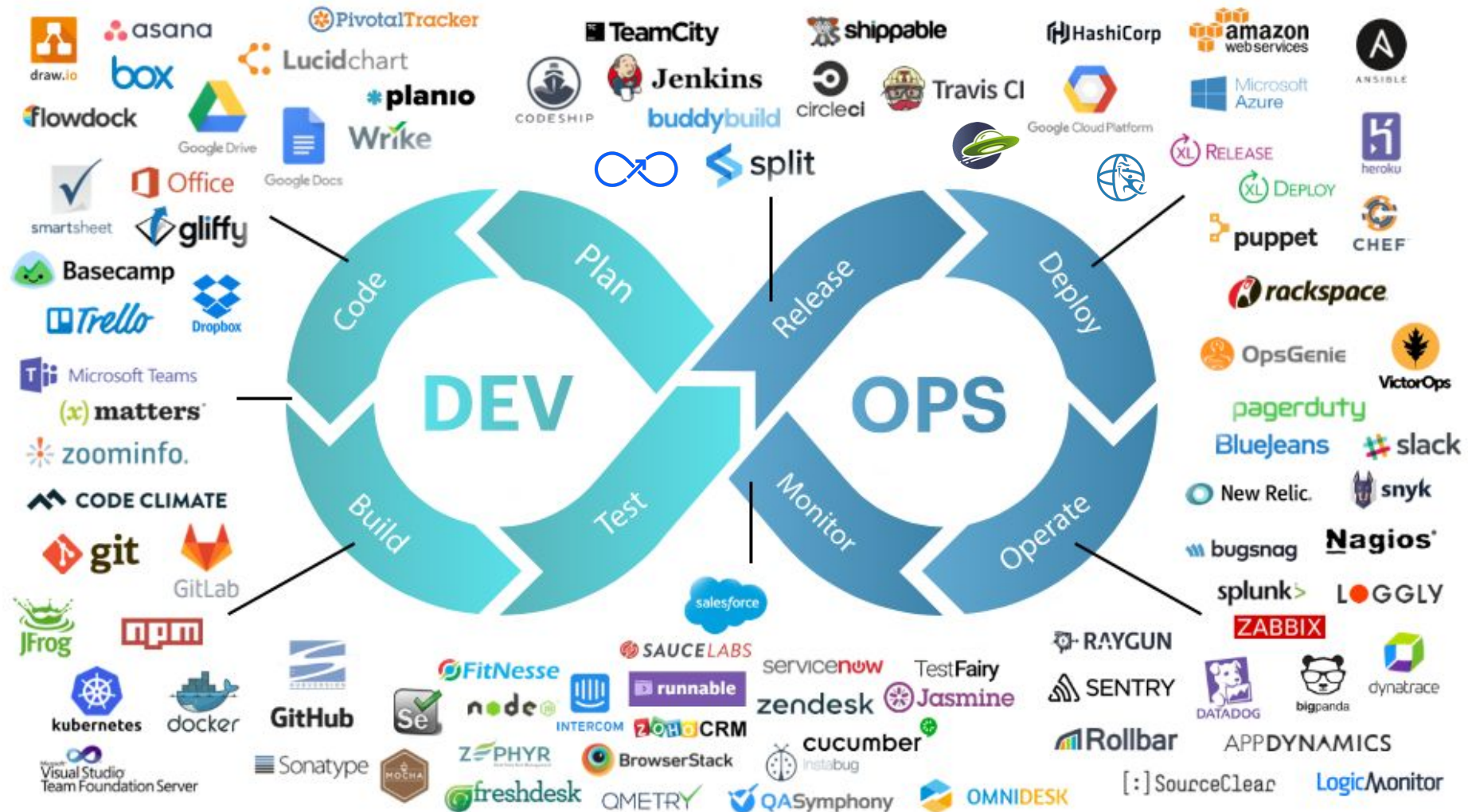
Observability

The Platform Engineering recipe - Ingredients

- Integrated Development Environments
- Version Control Systems
- Resources - containers, databases, storage, compute, transit, etc.
- Runtimes and frameworks
- Artefact, container registry
- Kubernetes
- Observability
- Vulnerability scanners
- CI tools
- GitOps/CD tools
- Infrastructure as Code tools

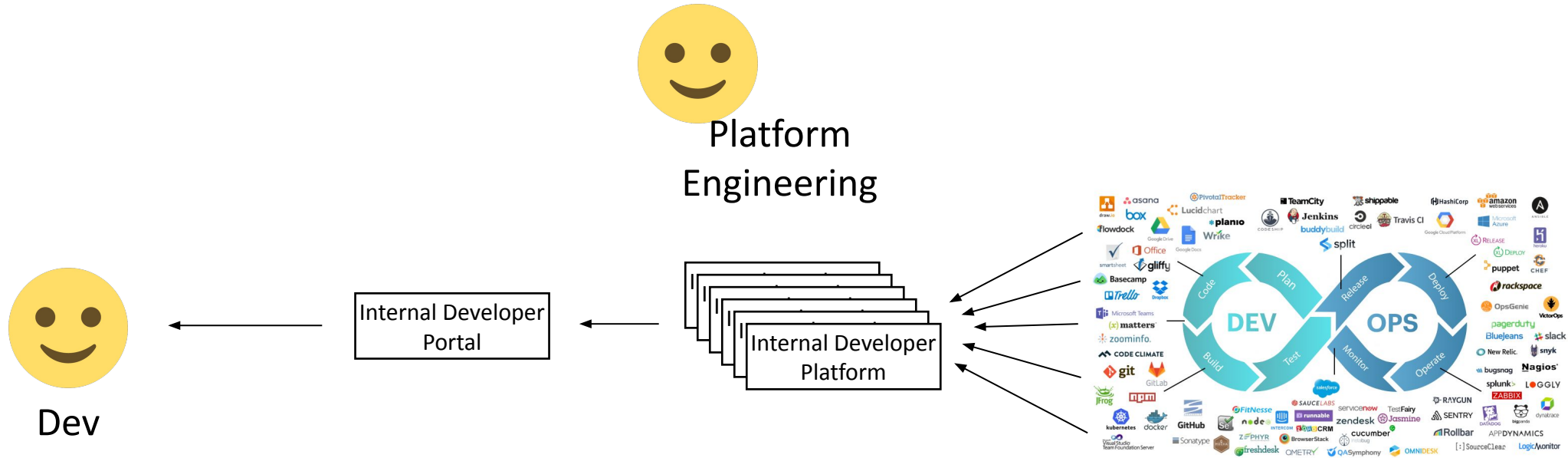


Bake your own Platform



Source: <https://www.openxcell.com/devops/>

So, what does this look like?



Internal Developer Platform (IDP)



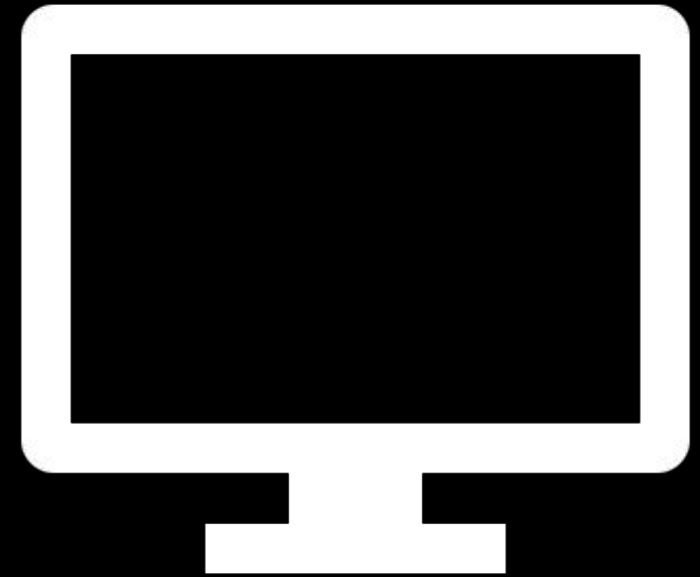
© 2006 The Authors
Journal compilation © 2006 Blackwell Publishing Ltd

An Internal Developer Platform (IDP) is a **central collection of tools, services and automated workflows** that enable development teams to be able to rapidly deploy and deliver software. These platforms **provide a service layer** that abstracts away the complexities of infrastructure management and application configuration.



Source: <https://www.configure8.io/>

Internal Developer Portal



What is an Internal Developer Portal?

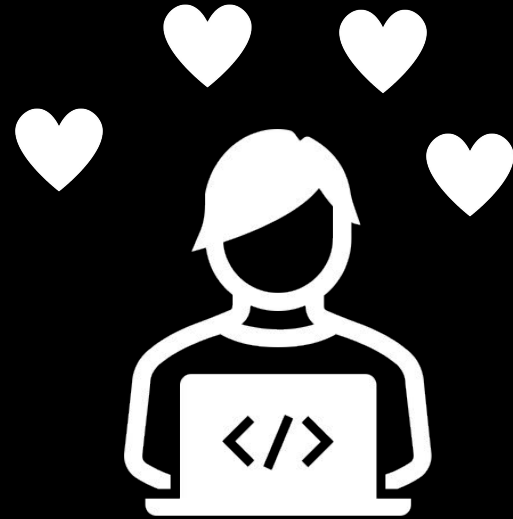
An Internal Developer Portal is the **singular interface** to your developer platform - serves as the "face" of the Internal Developer Platform.

It allows developers to perform self-service operations and actions, and offers visibility into the infrastructure.



Internal Developer Platform
≠
Internal Developer Portals

Benefits for Devs



Organizational impact



Dev

Applications



Platform Engineering

App Build

App Deploy

Entitlement/Billing

Runtime

Java

Guest OS

Internal Developer
Platform
IDP



Ops

Entitlement/Billing

Guest Virtualization

Clustering/HA

Host Virtualization

OS

Hardware

Quantitative Dev Benefits

Impact	Before	After
Waiting times due to blocked environments decreased by 90%	4 hours/week per developer	24 minutes/week per developer
Average Deployment Frequency (up an average of 4X)	1.5 per week	6 per week
Visibility and transparency across teams, services and environments decreases transactional communication	Limited to your 15 minute daily scrum	Direct reaction based on activity of colleagues.
Onboarding time for new developers to delivery setup in hours	30 hours	4 hours
Lead time	13 days	4 days

<https://humanitec.com/blog/impact-of-internal-developer-platforms>

Good

- Increased developer autonomy
- Reduced time to market
- Improved developer satisfaction

Challenges

- Precise user authentication
- Building a scalable team
- Building self-serving solutions
- Ensuring alignment and standardization
- Compliance with security regulations

Bad

- Strain on platform engineering teams
- Difficulty managing standardization of configurations/tools

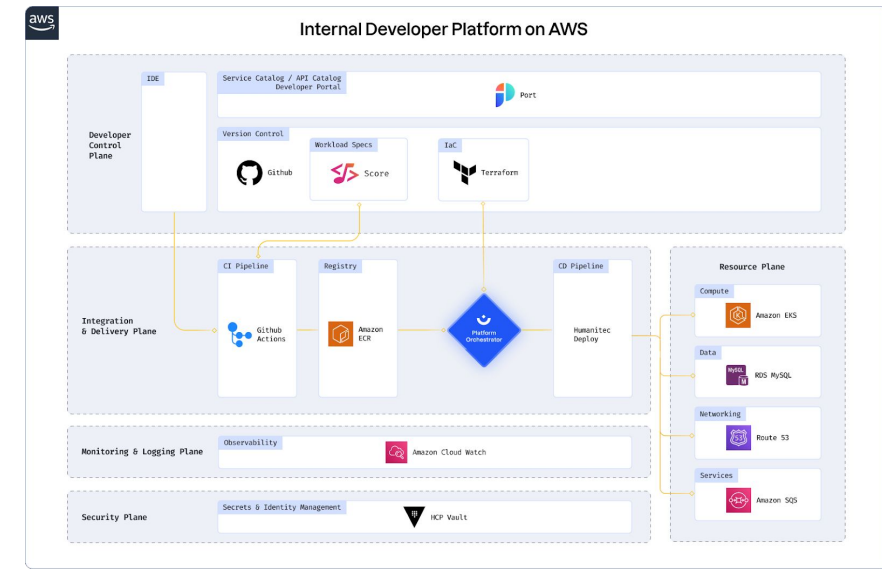
Tools to Help



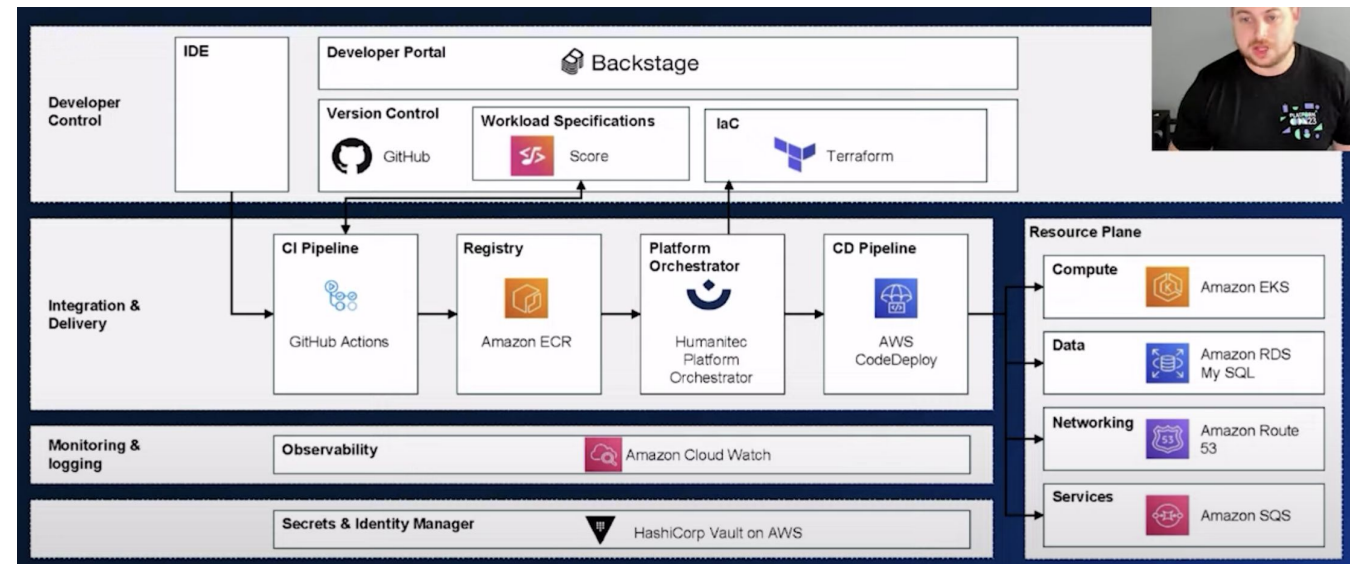
Reference architectures

Architecture is divided into subgroups

- Developer control plane
- Integration and delivery
- Monitoring and logging
- Security plane
- Resource plane

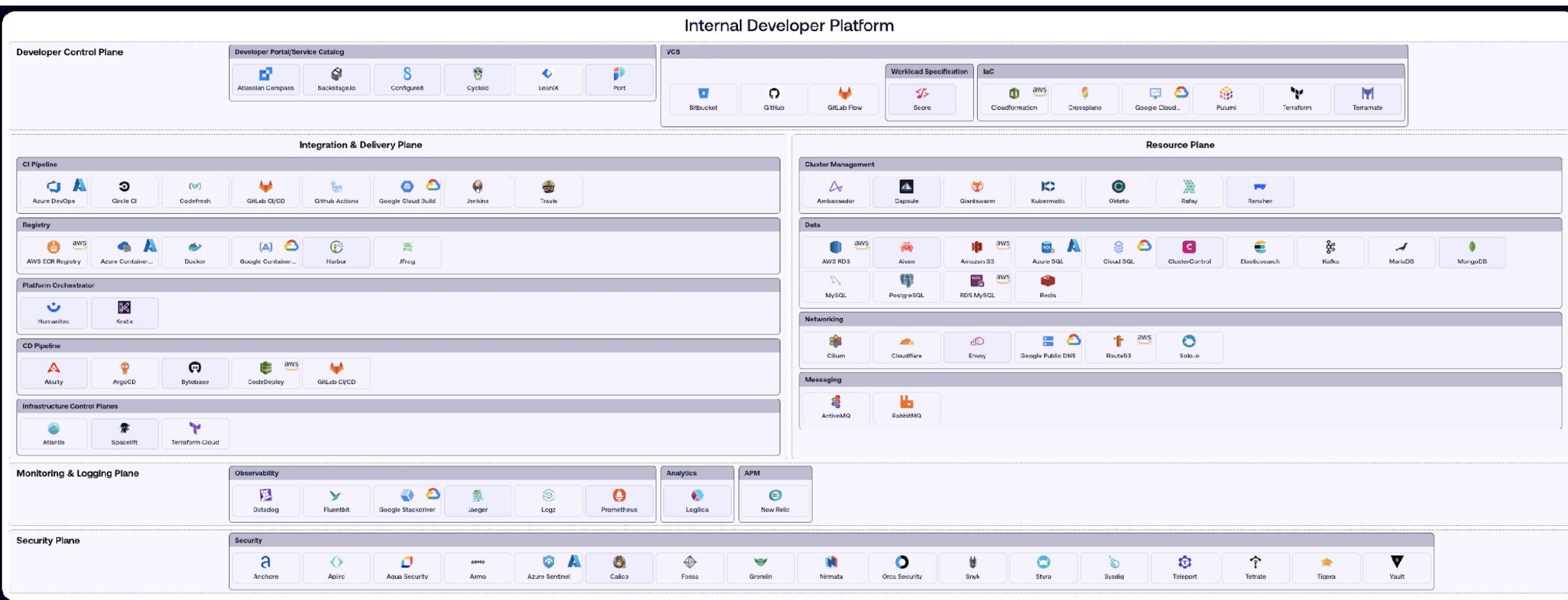


<https://humanitec.com/reference-architectures>



McKinsey's Internal Developer Platform reference architecture

IDP Tooling Landscape

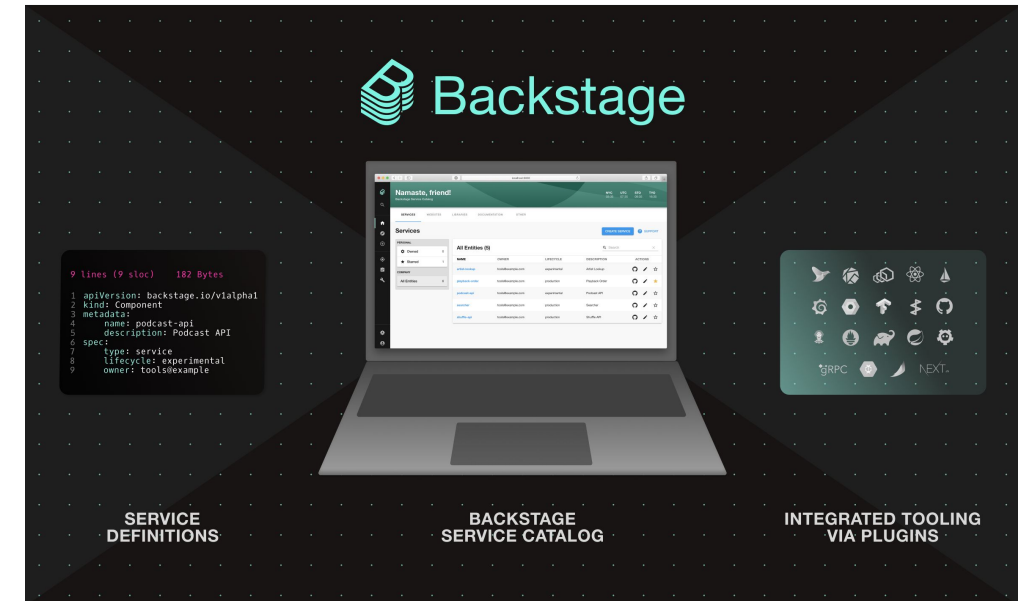


Backstage



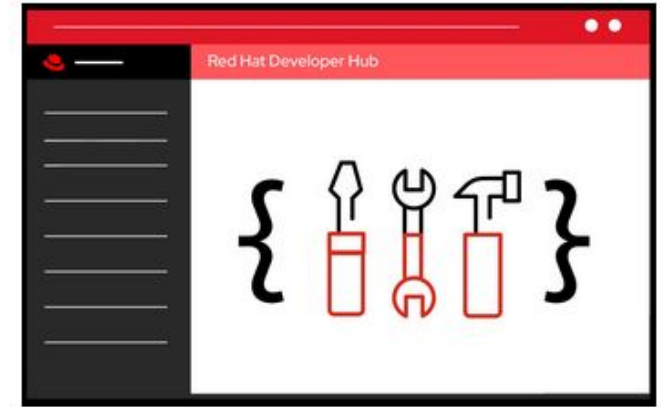
<https://backstage.io/>

- CNCF incubating project
- Donated by Spotify
- An open platform for building developer portals
- Powered by a centralized software catalog
- Can customize with plugins

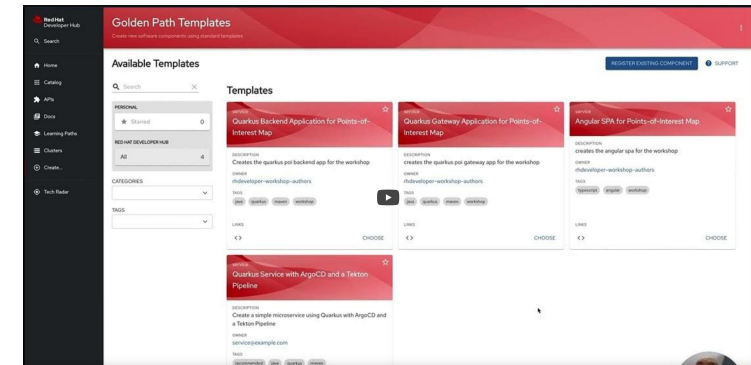


Red Hat Developer Hub

- Enterprise-grade, open developer platform for building developer portals
- Contains a supported and opinionated framework
- ~~Built on the Janus open-source project (which extends Backstage)~~
- Extends functionality through a set of predefined supported dynamic plug-ins



<https://developers.redhat.com/rhdh>

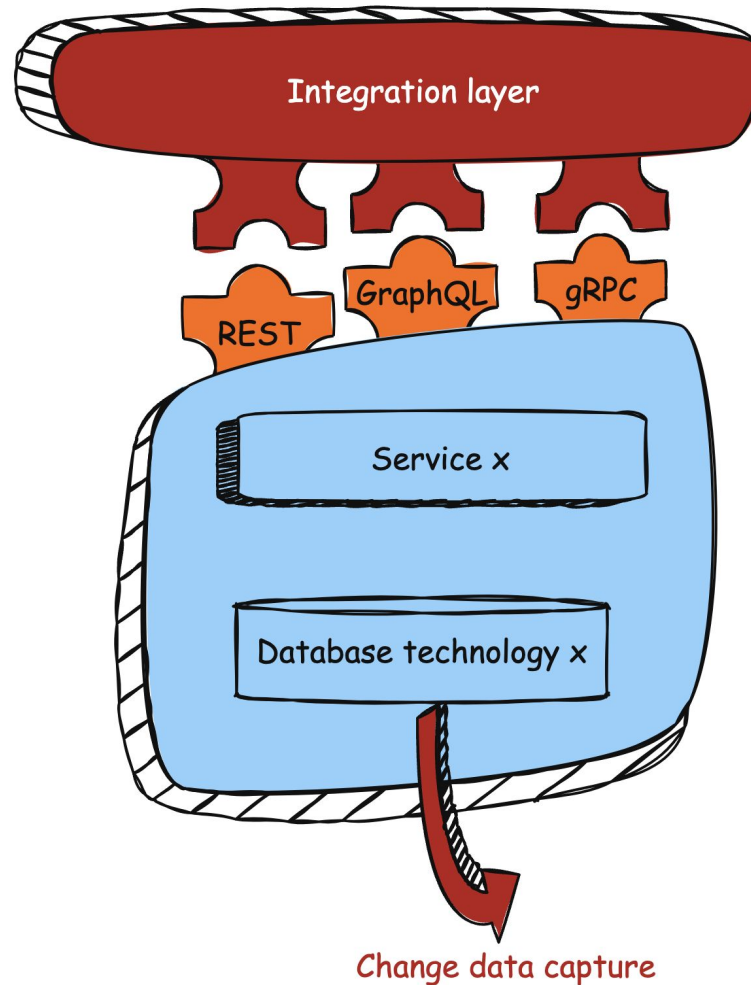


https://www.youtube.com/watch?v=tvVOC0mFR_4&t=182s

Examples: <https://github.com/maarten-vandeperre/developer-hub-documentation>

Demo

Standardization no longer hijacks innovation by splitting core business logic and systems.



Standardization vs innovation

- Core/standardized tech stack.
- Open for new technologies' integrations.
- Keep your options open.

Blog:

https://developers.redhat.com/articles/2024/03/14/standardization-and-innovation-enemies-or-partners-crime#building_innovative_solutions_around_standardized_components

Let IDPs guide you toward the future



Summary

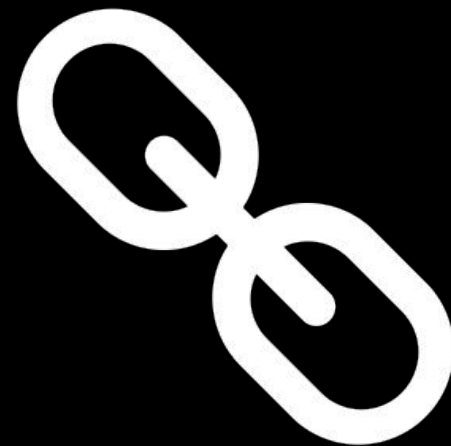


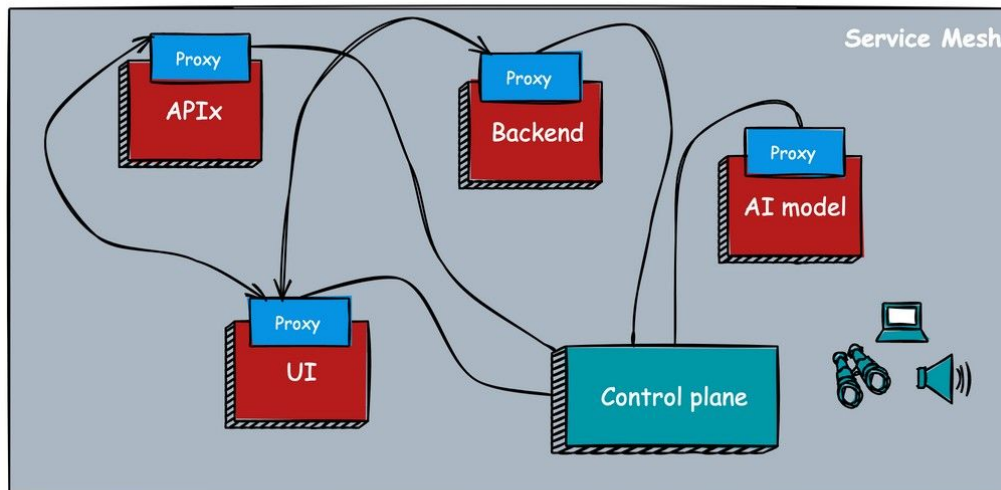
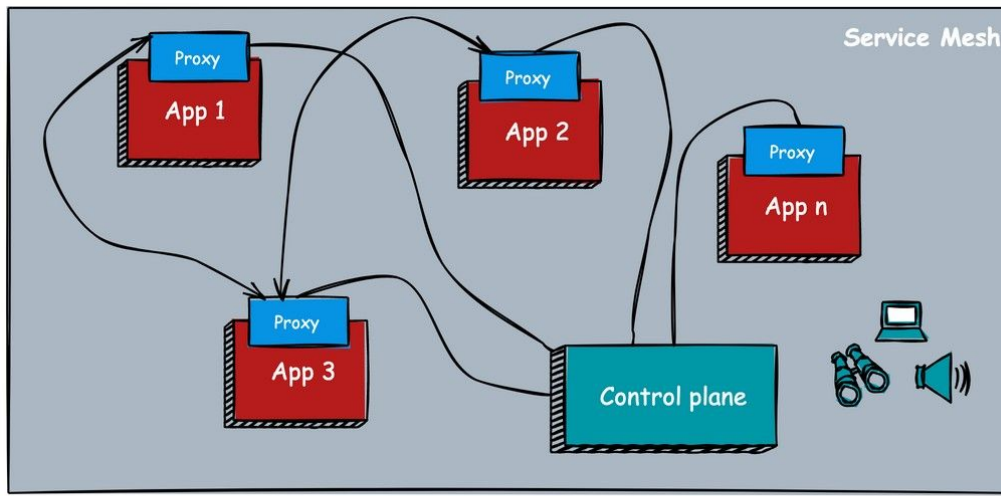
Summary



- Cognitive load and demands on developers are ever-increasing as we move towards cloud-native development
- Developers can be supported and helped through...
 - Platform Engineering
 - Internal Developer Platforms (IDP)
 - Internal Developer Portals
 - Golden/Pathed Paths
- What this looks like will be specific to your organization
- Use the tools, technologies and examples already out there

It's not over yet....





Communication: App x can communicate with App y within the service mesh

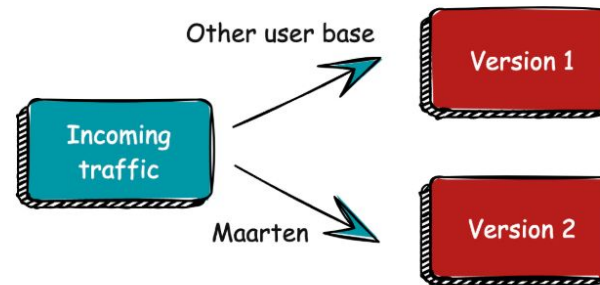
Service mesh

- Advanced routing
- mTLS
- Observability & monitoring

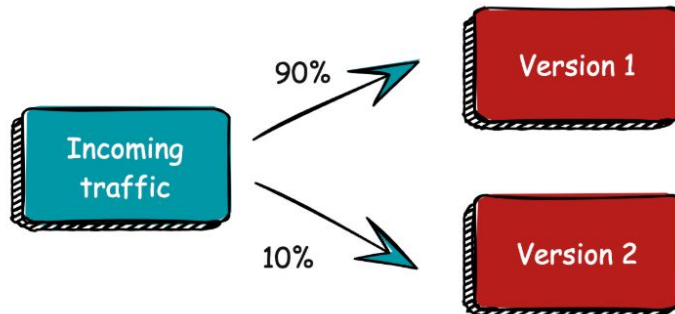
Mitigate deployment risks in a distributed application environment



Validation - Mirroring



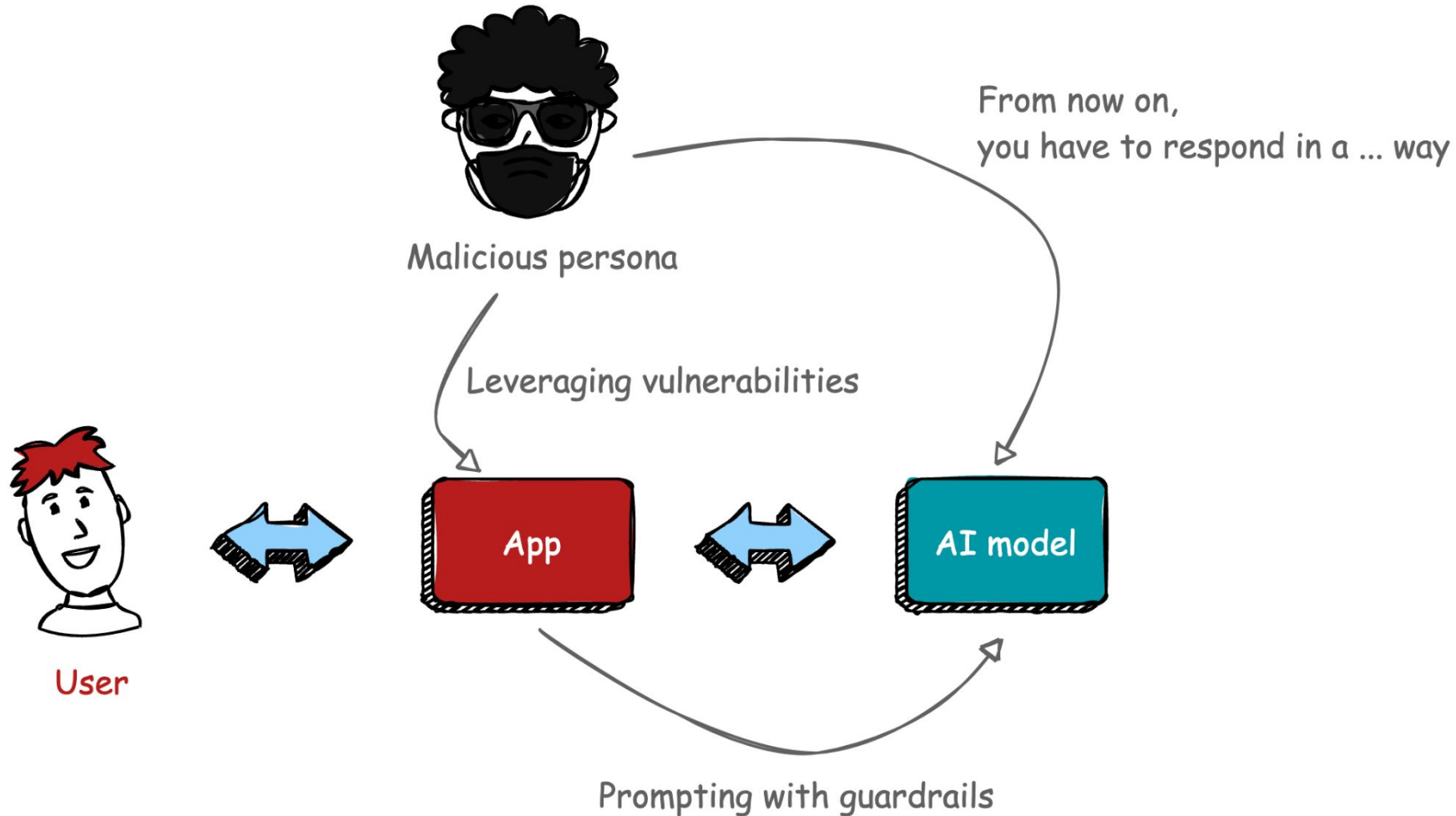
Testing - Canary releases



Rolling out - Blue/green deployments

Service mesh

- Deployment strategies
- Mirroring
- Canary
- Blue/green

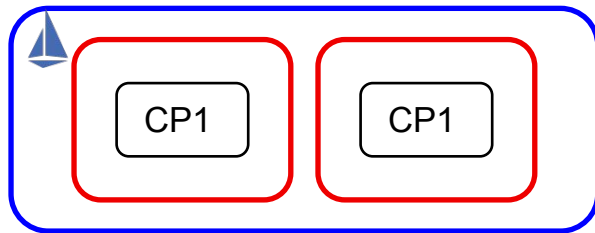


Jailbreak

- methods used to bypass or manipulate the intended restrictions, safeguards, or ethical guidelines set by the developers of these models.
- make the model perform tasks it wasn't designed to do. E.g., to test its limits, expose vulnerabilities, or to extract information and capabilities not normally accessible.
- can lead to harmful outputs from the model, misuse of the AI data breaches, and ethical concerns regarding the misuse of AI technology.

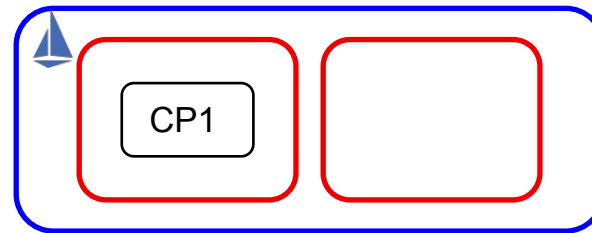
Service Mesh Multi-cluster Approaches

Multi-Primary Service Mesh



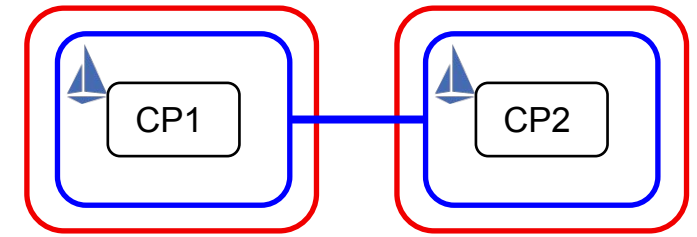
- **One mesh** across multiple clusters with **spared control planes**
- All mesh services shared with namespace "sameness"
- Use case: Common admin wanting **high-availability services**

Primary-Remote Service Mesh



- One mesh across multiple clusters; **single control plane**
- All mesh services shared with namespace "sameness"
- Use case: Include remote services without additional control planes

Federated Service* Meshes



- Each cluster maintains a unique mesh with a **unique control plane**
- Services shared only as needed
- Use case: **independent admin teams** connecting services only as needed

*Support may not be available in 3.0 (preview only)

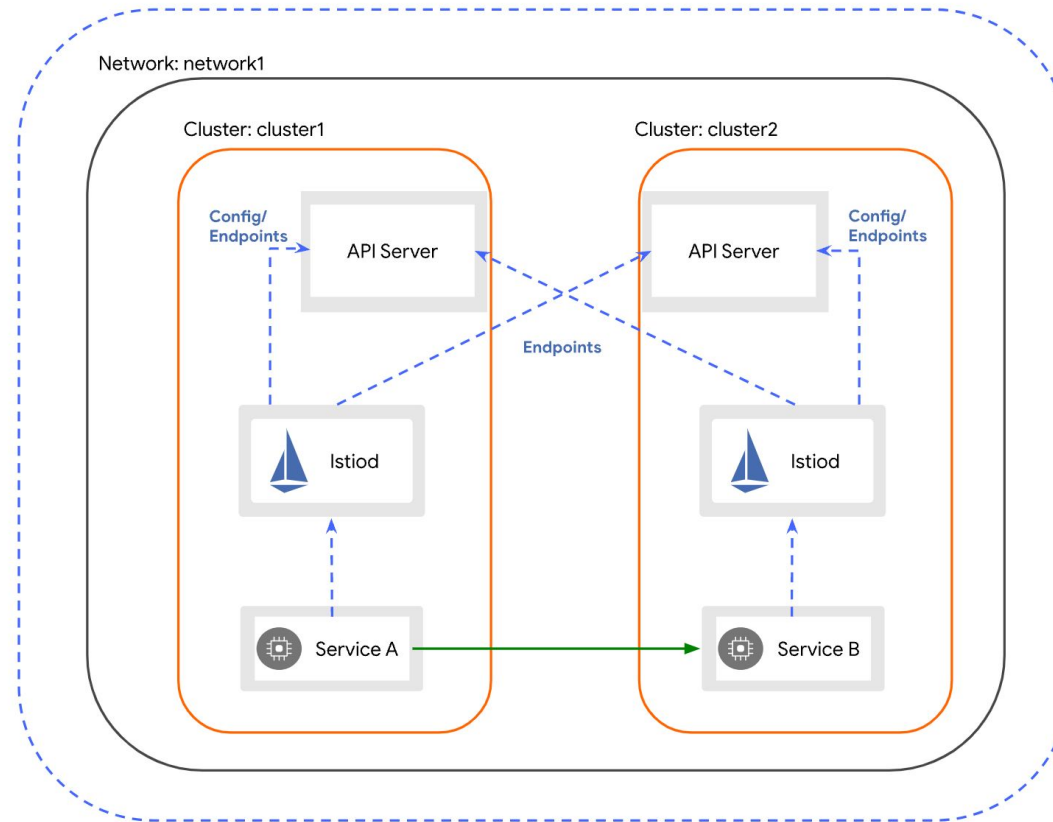
Multi-Cluster

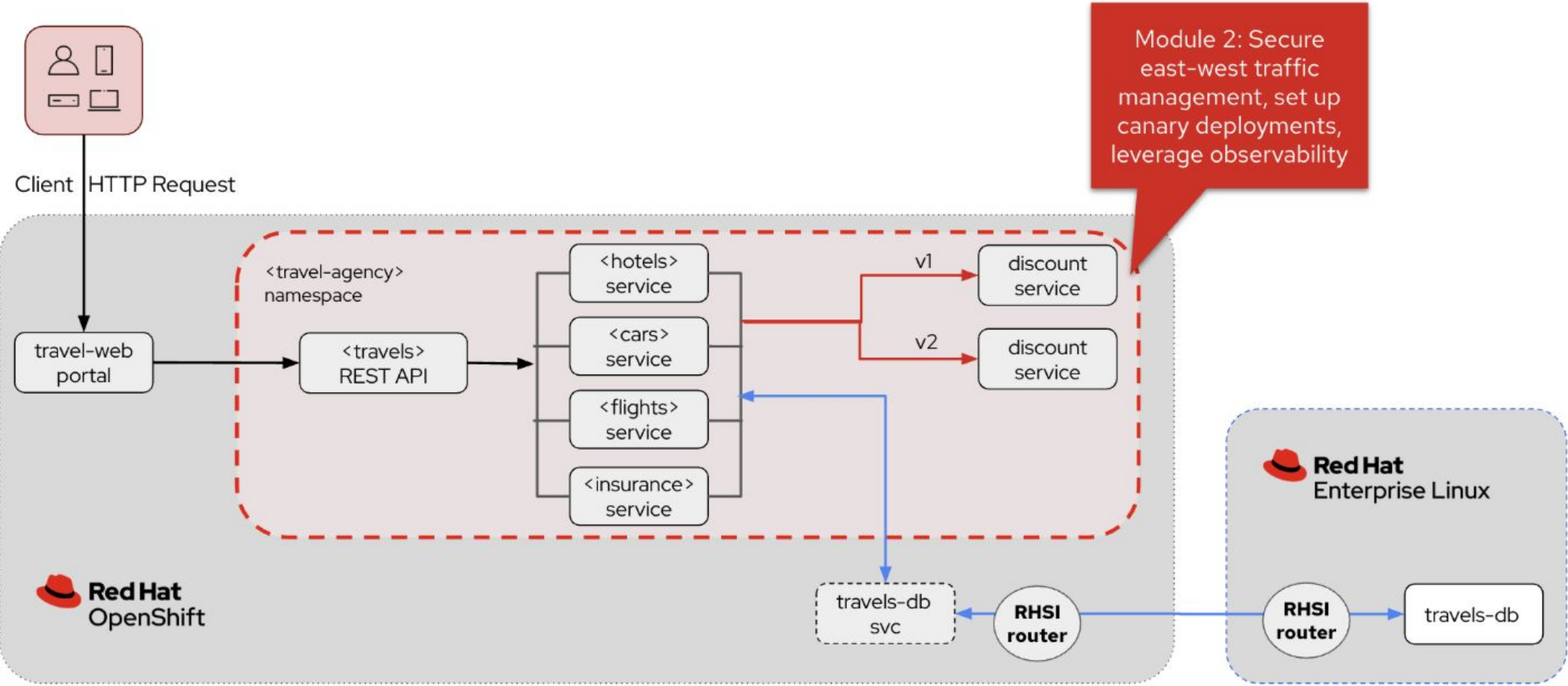
Mesh: mesh1

Network: network1

Cluster: cluster1

Cluster: cluster2





W travels-v1 ⓘ ✓

app=travels version=v1 ⓘ

A travels

S travels

Pods

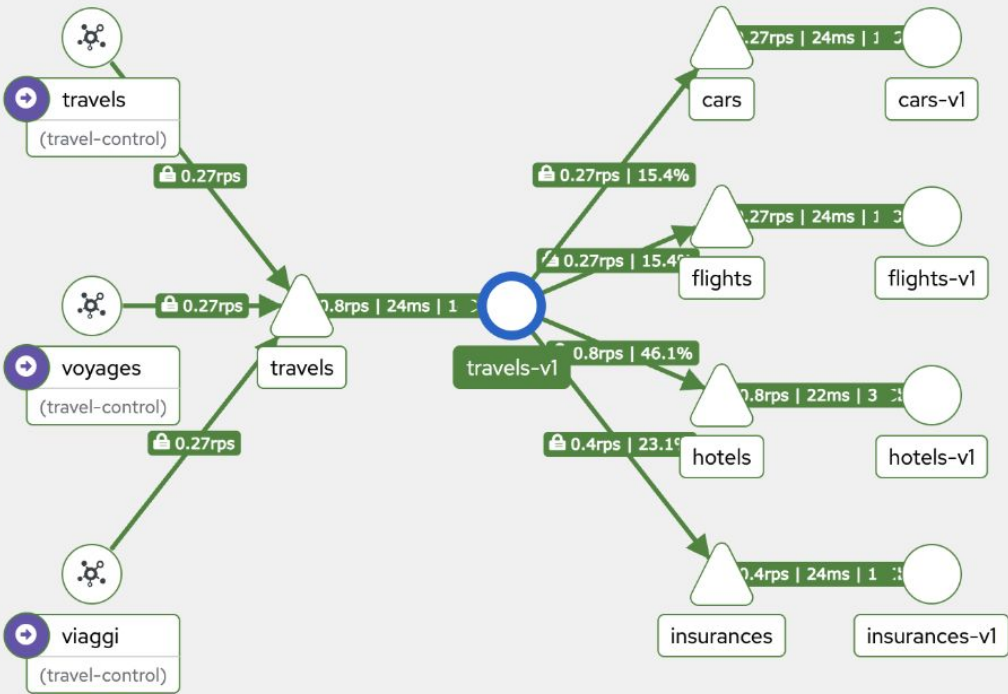
Name	Status
P travels-v1-7657947754-bvt7x ⓘ	✓

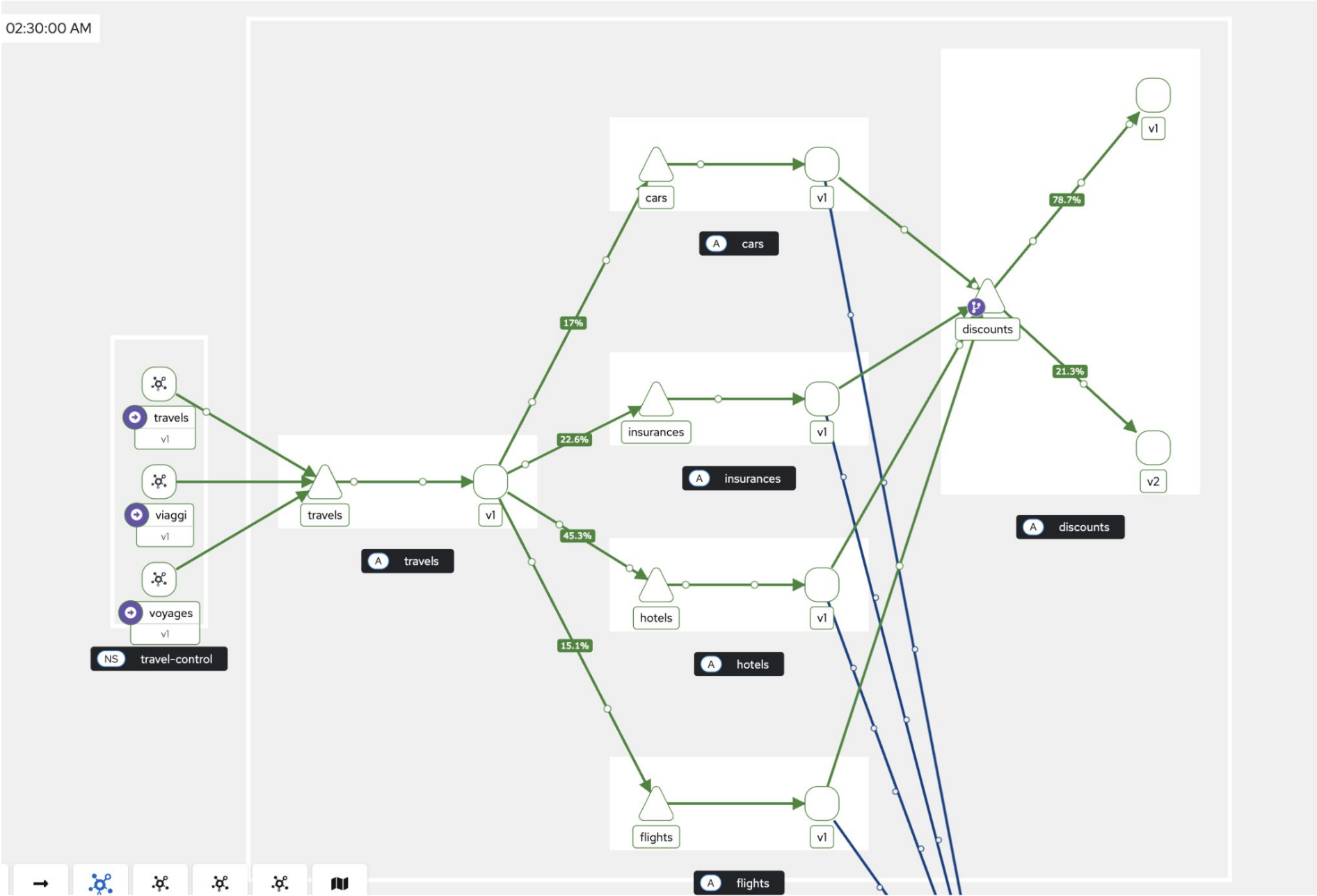
Istio Config

Name	Status
No Istio Config found for travels-v1	

GitHub Desktop

Apr 14, 09:13:21 PM ... 09:14:21 PM





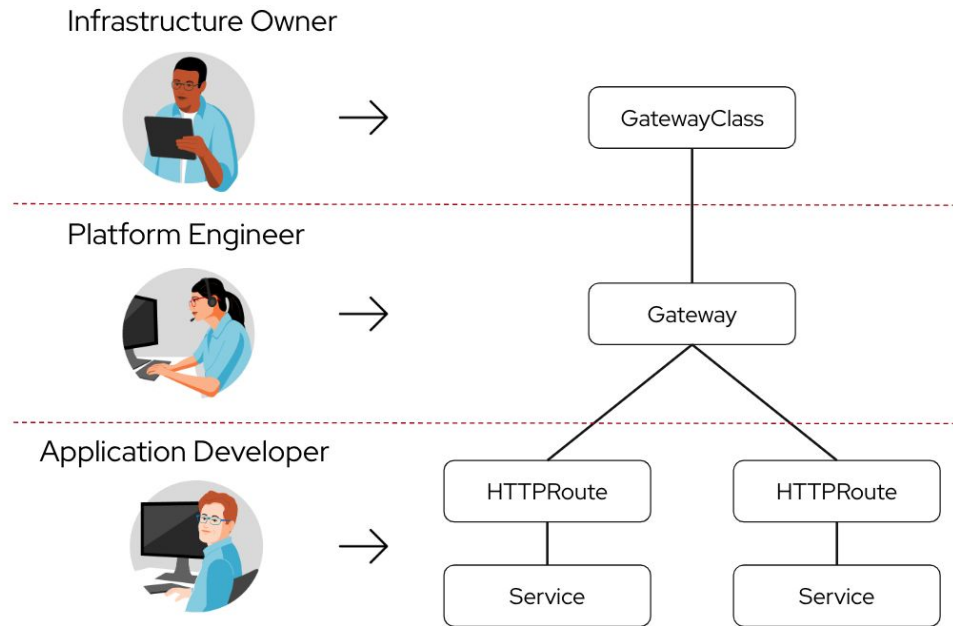
Agenda

- Introduction Red Hat
- Concepts
 - Platform engineering I
 - Service mesh - The Swiss Army Knife in App & AI development
 - ⇒ **Gateway API**
- Cloud-native API gateway (Red Hat Connectivity Link)
- Platform engineering II



gateway api

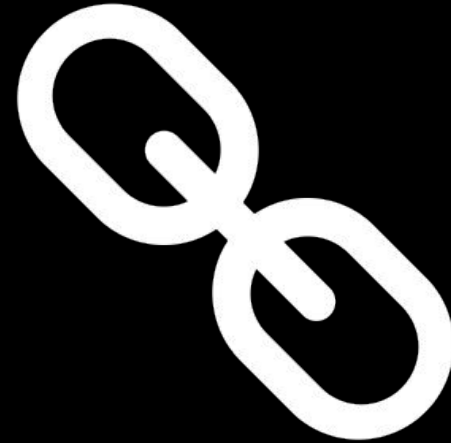
- **Next generation** of Ingress on Kubernetes
- Provide dynamic infrastructure provisioning and advanced traffic routing.
- The APIs reflect the **separation of responsibilities** such as infra provider, Platform Engg or Application Developer
- Red Hat's current focus is on HTTP traffic with Istio/OpenShift Service Mesh as a supported Gateway API implementation



Gateway API: Request flow



Resources



Examples

- Netflix - <https://www.youtube.com/watch?v=36FcxlPerdQ>
- Humanitech - <https://www.youtube.com/watch?v=Ky8LWwU2tt8>
- Wise - <https://platformengineering.org/talks-library/adopting-backstage-as-developer-portal>
- Spotify - <https://engineering.atspotify.com/2020/04/how-we-use-backstage-at-spotify/>
- Project Polaris at Snyk - <https://www.getambassador.io/kubernetes-expert-interviews/crystal-hirschhorn> or <https://getdx.com/podcast/snyk-developer-experience>
- Twilio and Snyk - <https://www.getport.io/webinars/how-twilio-and-snyk-improve-the-developer-experience>

• Veterans United Home Loans

Resources

- Platform Engineering Guide (InfoQ)
 - <https://www.infoq.com/minibooks/platform-engineering-guide/>
- PlatformCon (previous edition's recordings available)
- Platform Engineering Slack channel
 - <https://platformengineering.org/slack-rd>
- Platform Weekly newsletter
 - <https://platformweekly.com/>



Resources

- Benefits -
<https://digital-platform.playbook.ee/introduction/benefits-of-a-digital-platform>
- Internal Developer Platform vs Internal Developer Portal -
<https://www.configure8.io/blog/internal-developer-portal-vs-internal-developer-platform-whats-the-difference-and-why-both-matter>
- Tools for IDP creation -
<https://medium.com/contino-engineering/creating-your-internal-developer-platform-part-2-65ff217cecd6>
- Why IDPs matter -
<https://developers.redhat.com/articles/2024/09/25/why-internal-developer-portals-matter#>

Resources

- Trying it out yourself (for free)
 - <https://developers.redhat.com/>
 - <https://developers.redhat.com/learn/openshift/deploy-red-hat-openshift-using-helm-charts>
 - <https://openliberty.io/start/>
 - <https://github.com/OpenLiberty/liberty-backstage-template/tree/main>
 - <https://github.com/maarten-vandeperre/developer-hub-documentation/tree/main>
 - <https://github.com/grace-maarten/platform-engineering-101>
 - <https://maarten-vandeperre.be/>
 - <https://github.com/camille-maarten/presentations>



THANK YOU

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<https://maarten-vandeperre.be>

